C 40606 (Pages : 4) Name.....

Reg. No.....

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2023

(CBCSS—UG)

Mathematics

MTS 6B 13—DIFFERENTIAL EQUATIONS

(2019 Admission onwards)

Time : Two Hours and a Half Maximum : 80 Marks

Section A

Short Answer Type Questions.
Ceiling 25 *Marks.*

1. Find the solution of the differential equation $\frac{dp}{dt} = 0.5p - 450$.

2. Solve the differential equation $\left(4 + t^2\right)\frac{dy}{dt} + 2ty = 4t$.

3. State Existence and Uniqueness Theorem for First-Order Linear Differential Equations.

4. Solve the initial value problem $y' = y^2, \quad y(0) = 1$.

5. Find the general solution of $y'' + 5y' + 6y = 0$.

6. If y_1 and y_2 are two solutions of the differential equation, $y'' + p(x)y' + q(x)y = 0$.

   Show that $c_1y_1 + c_2y_2$ is also a solution for any values of the constants c_1 and c_2.

7. Let $y_1 = e^t \sin t, y_2 = e^t \cos t$. Find the Wronskian $W[y_1, y_2]$.

8. Solve the differential equation $y'' - 2y' - 3y = 3e^{2t}$.

9. Find the Laplace transform of e^{at}.

Turn over

10. Find $\mathcal{L}^{-1}\left(\frac{s}{s^2-a^2}\right)$ for s > | a |.

11. If $\mathrm{F}(s)=\mathcal{L}(f(t))$ exists for $s>a\geq 0$, and if c is a constant, Show that

$$\mathcal{L}\left(e^{ct}\,f(t)\right)=\mathrm{F}(s-c),\quad s>a+c.$$

12. Solve the boundary value problem $y''+y=0,\ y(0)=1, y(\pi)=a,$

where a is a given number.

13. Find the fundamental period of the function $\sin(5x)$.

14. Define an odd function. Prove that if $f(x)$ is an odd function then

$$\int_{-\mathrm{L}}^{\mathrm{L}} f(x)\,dx=0.$$

15. Verify that the method of separation of variables may be used to solve the equation $xu_{xx}+u_t=0$.

(2 Marks each)

Section B

Paragraph / Problem Type Questions.
Ceiling 35 *Marks.*

16. Show that the equation

$$\frac{dy}{dx}=\frac{x^2}{1-y^2}.$$

is separable, and then find an equation for its integral curves.

17. Solve the differential equation

$$\left(y\cos x+2xe^y\right)+\left(\sin x+x^2e^y-1\right)y'=0.$$

18. Given that $y_1(t) = t^{-1}$ is a solution of

$$2t^2y'' + 3ty' - y = 0, t > 0,$$

find a fundamental set of solutions.

19. Find the general solution of the differential equation $y'' + y = \tan t$ on $0 < t < \pi/2$.

20. Find the Laplace transform of the following function $f(t) = \int_0^t (t-\tau)^2 \cos(2\tau)\, d\tau$.

21. Find the inverse Laplace transform of the following function using the convolution theorem

$$F(s) = \frac{1}{(s+1)^2(s^2+4)}.$$

22. Determine the co-efficients in the Fourier series of the function

$$f(x) = \begin{cases} -x, & -2 \le x \le 0, \\ x, & 0 \le x \le 2 \end{cases}$$

with $f(x+4) = f(x)$.

23. Find the displacement $u(x,t)$ of the vibrating string of length $L = 30$ that satisfies the wave equation

$$4u_{xx} = u_{tt}, \quad 0 < x < 30, t > 0.$$

Assume that the ends of the string are fixed and that the string is set in motion with no initial velocity from the initial position

$$u(x,0) = f(x) = \begin{cases} x/10, & 0 \le x \le 10, \\ (30-x)/20, & 10 < x \le 30 \end{cases}.$$

(5 marks each)

Turn over

Section C (Essay Type Questions)

two *out of four.*

24. (a) Find the general solution of the differential equation $\frac{dy}{dt} - 2y = 4 - t$ by the method of integrating factors.

(b) Find the value of b for which the following equation is exact, and then solve it using that value of b

$$\left(ye^{2xy} + x\right) + bxe^{2xy}y' = 0.$$

25. Find a series solution in powers of x of Airy's equation

$$y'' - xy = 0, \quad -\infty < x < \infty.$$

26. Use the Laplace transform and solve the following initial value problem

$$y'' + 3y' + 2y = 0\,; \quad y(0) = 1, y'(0) = 0.$$

27. Find the Fourier series of the following periodic function $f(x)$ of period $p = 2\mathrm{L}$ defined by

$$f(x) = 3x^2 \quad -1 < x < 1.$$

(10 marks each)

C 40606-A (Pages : 5) Name.....

Reg. No.....

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2023

(CBCSS—UG)

Mathematics

MTS 6B 13—DIFFERENTIAL EQUATIONS

(2019 Admission onwards)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes	Total No. of Questions : 20	Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 6B 13—DIFFERENTIAL EQUATIONS

(Multiple Choice Questions for SDE Candidates)

1. Solution of $\left(1+4xy+2y^2\right)dx+\left(1+4xy+2x^2\right)dy=0$ is ————————.

   (A) $x+2x^2y+2xy^2+y=c$ where c is an arbitrary constant.

   (B) $x-2x^2y+2xy^2+y=c$ where c is an arbitrary constant.

   (C) $x+2x^2y-2xy^2+y=c$ where c is an arbitrary constant.

   (D) $x+2x^2y+2xy^2-y=c$ where c is an arbitrary constant.

2. The electrostatic potential $v=v\left(r\right)$ between two concentric spheres of radii $r_1=5$ cm and $r_2=10$ cm kept at potentials $v_1=110$V and $v_2=0$, respectively is given by ————————.

   (A) $v\left(r\right)=-110+1100/r$ V. (B) $v\left(r\right)=1100/r$ V.

   (C) $v\left(r\right)=-110+1100/r^2$ V. (D) None of the above options.

3. Solution of $y''-y'-2y=10\cos t$ is ————————.

   (A) $y=c_1e^{-t}+c_2e^{2t}-3\cos t-\sin t.$ (B) $y=c_1e^{-2t}+c_2e^{2t}-3\cos t-\sin t.$

   (C) $y=c_1e^{-t}+c_2e^{2t}-3\cos 2t-\sin t.$ (D) $y=c_1e^{-t}+c_2e^{2t}-3\cos t-\sin 4t.$

4. A particular solution of $y''-3y'-4y=3e^{2t}+2\sin t-8e^t\cos 2t$ is ————————.

   (A) $\frac{1}{4}e^{2t}+\frac{3}{17}\cos t-\frac{5}{17}\sin t+\frac{10}{13}e^t\cos 2t+\frac{2}{13}e^t\sin 2t.$

   (B) $-\frac{1}{4}e^{2t}+\frac{3}{17}\cos t-\frac{5}{17}\sin t+\frac{10}{13}e^t\cos 2t+\frac{2}{13}e^t\sin 2t.$

   (C) $\frac{1}{2}e^{2t}+\frac{3}{17}\cos t-\frac{5}{17}\sin t+\frac{10}{13}e^t\cos 2t+\frac{2}{13}e^t\sin 2t.$

   (D) $-\frac{1}{2}e^{2t}+\frac{3}{17}\cos t-\frac{5}{17}\sin t+\frac{10}{13}e^t\cos 2t+\frac{2}{13}e^t\sin 2t.$

5. Solution of $y''-4y'+3y=\sin 3x\cos 2x$ is ————————.

   (A) $c_1e^x+c_2e^{3x}+\frac{10}{884}\cos 5x-\frac{11}{884}\sin 5x+\frac{1}{10}\cos x+\frac{1}{2}\sin x.$

   (B) $c_1e^x+c_2e^{3x}+\frac{10}{884}\cos 5x-\frac{11}{884}\sin 5x+\frac{1}{10}\cos x+\frac{1}{20}\sin x.$

   (C) $c_1e^x+c_2e^{3x}+\frac{10}{884}\cos 5x-\frac{11}{884}\sin 5x-\frac{1}{10}\cos x+\frac{1}{20}\sin x.$

   (D) $c_1e^x+c_2e^{3x}+\frac{10}{884}\cos 5x-\frac{11}{884}\sin 5x+\frac{1}{10}\cos x-\frac{1}{20}\sin x.$

6. Solution of $(D-1)^2 y = t + e^t$ is ————————.

(A) $(c_1 + c_2 t) e^t + t + 2 + \frac{1}{2} t^3 e^t$.

(B) $(c_1 + c_2 t) e^t + t + 2 + \frac{1}{2} t^2 e^{2t}$.

(C) $(c_1 + c_2 t) e^t + t + 2 + \frac{1}{2} t^2 e^t$.

(D) $(c_1 + c_2 t) e^t + t + 2 + \frac{1}{3} t^2 e^t$.

7. The particular solution of $ty'' - y' = (3+t) t^2 e^t$ is ————————.

(A) $-\frac{1}{5}\int (3+t) t^2 e^t + \frac{t^2}{2}\int (3+t) e^t \, dt$.

(B) $\frac{1}{5}\int (3+t) t^2 e^t + \frac{t^2}{2}\int (3+t) e^t \, dt$.

(C) $-\frac{1}{2}\int (3+t) t^2 e^t + \frac{t^2}{2}\int (3+t) e^t \, dt$.

(D) $\frac{1}{2}\int (3+t) t^2 e^t + \frac{t^2}{2}\int (3+t) e^t \, dt$.

8. The Laplace transform of t is ————————.

(A) $\frac{1}{s}$, if $s > 0$.

(B) $\frac{1}{s^2}$, if $s > 0$.

(C) $-\frac{1}{s^2}$, if $s > 0$.

(D) None of the above options.

9. Which one of the following statements is true ?

(A) If a and b are any two constants and $f(t)$ and $g(t)$ are functions with Laplace transforms $F(s)$ and $G(s)$ respectively, then $L[a f(t) + b g(t)] = aF(s) + bG(s)$.

(B) If a and b are any two constants and $f(t)$ and $g(t)$ are functions with Laplace transforms $F(s)$ and $G(s)$ respectively, then $L[a f(t) + b g(t)] = aF(s) - bG(s)$.

(C) If a and b are any two constants and $f(t)$ and $g(t)$ are functions with Laplace transforms $F(s)$ and $G(s)$ respectively, then $L[a f(t) + b g(t)] = -aF(s) + bG(s)$.

(D) If a and b are any two constants and $f(t)$ and $g(t)$ are functions with Laplace transforms $F(s)$ and $G(s)$ respectively, then $L[a f(t) + b g(t)] = -aF(s) - bG(s)$.

Turn over

10. Laplace transform of $\sinh at$ is _____.

(A) $\frac{s}{s^2 + a^2}$. (B) $\frac{s}{s^2 - a^2}$.

(C) $\frac{a}{s^2 - a^2}$. (D) $\frac{a}{s^2 + a^2}$.

11. The inverse Laplace transform of $\frac{5}{s+3}$ is :

(A) $5e^{-3t}$. (B) $-5e^{-3t}$.

(C) $5e^{-5t}$. (D) $-5e^{-5t}$.

12. $L^{-1}\left[\frac{s}{s^2 + 25}\right] =$ _____.

(A) $-5\cos 5t$. (B) $\cos 4t$.

(C) $-\cos 5t$. (D) $\cos 5t$.

13. $L^{-1}\left[\frac{1}{s^2 + 81}\right] =$ _____.

(A) $-\frac{1}{9}\sin 9t$. (B) $\frac{1}{9}\sin 9t$.

(C) $\frac{1}{10}\sin 9t$. (D) $-\frac{1}{10}\sin 9t$.

14. The inverse Laplace transform of $\frac{s+1}{s^2+1}$ is _____.

(A) $-\cos t - \sin t$. (B) $-\cos t + \sin t$.

(C) $\cos t - \sin t$. (D) $\cos t + \sin t$.

15. $L^{-1}\left[\frac{b_1}{s} + \frac{b_2}{s^2} + \frac{b_3}{s^3} + \frac{b_4}{s^4}\right] =$

(A) $b_1 + b_2 t + \frac{t^2}{2!} b_3 + \frac{t^3}{3!} b_4$. (B) $-b_1 + b_2 t + \frac{t^2}{2!} b_3 + \frac{t^3}{3!} b_4$.

(C) $b_1 - b_2 t + \frac{t^2}{2!} b_3 + \frac{t^3}{3!} b_4$. (D) $b_1 + b_2 t + \frac{t^2}{2!} b_3 - \frac{t^3}{3!} b_4$.

16. The inverse Laplace transform of $\frac{1}{s(s+1)(s+2)}$ is _____.

(A) $-\frac{1}{2}-e^{-t}+\frac{1}{2}e^{-2t}$. (B) $\frac{1}{2}-e^{-t}+\frac{1}{2}e^{-2t}$.

(C) $\frac{1}{2}-e^{-t}-\frac{1}{2}e^{-2t}$. (D) None of the above options.

17. The inverse Laplace transform of $\frac{1}{s^2}\left(\frac{s+1}{s^2+a}\right)$ is _____.

(A) $\frac{1}{a}\left(1+t-\cos\sqrt{a}\,t-\frac{\sin\sqrt{a}\,t}{\sqrt{a}}\right)$. (B) $-\frac{1}{a}\left(1+t-\cos\sqrt{a}\,t-\frac{\sin\sqrt{a}\,t}{\sqrt{a}}\right)$.

(C) $\frac{1}{a}\left(1+t-\cos\sqrt{a}\,t+\frac{\sin\sqrt{a}\,t}{\sqrt{a}}\right)$. (D) None of the above options.

18. The inverse transform of $\frac{s+2}{(s+2)^2+1}$ is _____.

(A) $e^{-2t}\cos t$. (B) $e^{2t}\cos t$.

(C) $e^{-2t}\sin t$. (D) $e^{2t}\sin t$.

19. $L^{-1}\left[\frac{s}{(s+3)^2+1}\right]=$ _____.

(A) $e^{-3t}\cos t-3e^{-3t}\sin t$. (B) $e^{3t}\cos t-3e^{-3t}\sin t$.

(C) $e^{-3t}\cos t+3e^{-3t}\sin t$. (D) $e^{-3t}\cos t-3e^{3t}\sin t$.

20. $L\left[t\,e^{at}\right]=$ _____.

(A) $\frac{1}{(s-a)^2}$, $s>a$. (B) $\frac{1}{(s+a)^2}$, $s>a$.

(C) $\frac{1}{(s-a)^3}$, $s>a$. (D) $\frac{1}{(s-a)}$, $s>a$.